

Varunprasaath R. Selvaraju, Ph.D.

(+1) 678-207-9914 | varunprasaath@gmail.com | linkedin.com/in/varunprasaath-rs |

PROFESSIONAL SUMMARY

PhD chemist/materials scientist with hands-on analytical characterization, LC/GC-MS-adjacent method experience, stability analysis, and DOE-guided process-development documentation. Connects structure-property data with technical reports and reproducible methods for cross-functional development teams.

CORE COMPETENCIES

Analytical Characterization Method Development & Validation LC/GC-MS Adjacent Analysis Impurity / Stability Analysis Process Development DOE/SPC

Technical Reports Cross-Functional Development

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead process-development and analytical characterization workflows for bio-based polymer systems, using DOE, spectroscopy, FT-IR, and thermal analysis to define reproducible material-quality readouts.
- Translate experimental results into technical reports, SOP-ready procedures, and process recommendations that support method consistency and scale-up decisions.
- Apply SPC/statistical thinking to identify critical process parameters and improve reproducibility across synthesis and characterization campaigns.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Developed structure-property datasets for 15+ organic and organometallic electrochemical materials using UV-Vis-NIR, FT-IR, NMR, GPC, cyclic voltammetry, spectroelectrochemistry, TGA, DSC, GC-MS, and LC-MS-adjacent methods.
- Investigated stability and degradation pathways through electrochemical and spectroscopic stress testing, supporting root-cause analysis for oxidative degradation, delamination, and redox instability.
- Maintained rigorous experimental notebooks, sample traceability, and method documentation for multi-technique analytical workflows.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded process development for high-performance molecular thin-film coatings, connecting process conditions with optical/material response through spectroscopy and thin-film characterization.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Developed QCM-D-monitored polymer thin-film deposition workflows under flow, using DOE-style process variation to assess growth kinetics and coating reproducibility.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed operations for a 15-member process-development laboratory, implementing equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30%.
- Trained researchers on instrument workflows, experimental documentation, and safe handling while coordinating multiple concurrent materials-development projects.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Multi-Technique Analytical Characterization Analytical Ops

Built structure-property datasets across functional materials using spectroscopic, electrochemical, chromatographic, and thermal techniques to assess stability and degradation. UV-Vis-NIR, FT-IR, LC-MS/GC-MS exposure, NMR, GPC, TGA, DSC

Stability and Degradation Root-Cause Analysis Impurity/Stability

Identified oxidative degradation, redox instability, and interface failure modes through stress testing and analytical readouts, translating findings into mitigation recommendations. Electrochemical stress testing, spectroscopy, root-cause analysis

DOE-Guided Process Development Method Reproducibility

Defined critical process parameters and reproducible characterization workflows for polymer process-development campaigns. DOE, SPC, SOPs, technical reports

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Analytical & Characterization: UV-Vis-NIR, FT-IR, NMR, GPC, GC-MS, LC-MS exposure, cyclic voltammetry, spectroelectrochemistry, TGA, DSC, QCM-D

Method Development: Experimental design, method documentation, reproducibility studies, validation mindset, sample traceability, technical reports

Process & Pharma-Adjacent: DOE, SPC, process windows, stability/degradation analysis, formulation/process-development adjacency, GMP documentation awareness

Data & Collaboration: Excel, Python/basic analysis, JMP/basic, cross-functional technical communication, risk assessment

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)